

Learn the Concept of Set Theory with Maxima

We have been experimenting with Maxima, a powerful Computer Algebraic System (CAS) for quite a while now in this series of articles. Maxima is a free and open source project that is continually being developed and improved upon. It evolved from Macsyma, which is now almost defunct. This is the 18th article in this series and it touches upon the fundamentals of set theory, through Maxima.



Most of you have definitely heard about set theory and may already know quite a bit about sets. Let us then go about getting the maths done by the computer. In this series, we have mostly been picking up mathematical concepts we're familiar with and then figuring out how we can work with these on the computer, with very little or no programming knowledge. The same holds true for sets. Let's get started with the fundamentals, starting with how they are created.

The creation of sets

A set is an unordered collection of distinct items—any item, in any order, but unique. An item is commonly referred to as an element. If an item is contained in a set, it is commonly referred to as a member of the set. A set is

typically represented by its members enclosed in braces $\{ \}$ and separated by commas. $\{6, -5, 9, 0\}$, $\{\text{dog, cat, donkey, cow, buffalo}\}$, $\{\text{Kiran, Kasturi, Karan}\}$, and $\{6, \text{horse, sapphire}\}$ are some examples. Notice that the first three sets have related items in them, but the last one doesn't. That's perfectly fine. However, if the items in a set have relation(s) or condition(s), the set can also be expressed with that relation(s) or condition(s) mentioned within braces $\{ \}$. For example, $\{\text{All human beings younger than 35 years}\}$, $\{\text{All positive even numbers}\}$, $\{\text{All multiples of 3}\}$. In Maxima, we can straight away represent the sets in the first notation, as follows:

```
S maxima -q
(%i1) {6, -5, 9, 0};
(%o1)
{6, -5, 9, 0}
```